Google Cloud H2S

Gen AI Academy

APAC Edition

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Participant Details

a. **Participant name: Zuha Jagirdar**
b. **Problem Statement:** Build and deploy a **single AI agent** using **ADK and Gemini** that is hosted on **Cloud Run** and performs **one clearly defined task**.
The agent must be callable via an HTTP endpoint and return a valid response for a given input.

Brief about the idea

The project focuses on building a **lightweight AI-based information retrieval agent** using Google ADK that can process user queries and return **accurate, structured, and concise responses**.

In many real-world applications, users require **quick factual information** rather than long conversational outputs. However, most existing AI systems are designed as general-purpose chatbots, which often produce verbose and unstructured answers.

To address this gap, this solution implements a **single-purpose AI agent** that prioritizes:

- clarity of response
- consistency in output format
- efficiency in processing

The agent leverages **Google Gemini for natural language understanding** and is deployed using **Google Cloud Run**, making it scalable, accessible, and suitable for real-time API-based applications.

This approach demonstrates how **focused AI systems can outperform complex architectures** when solving specific problems efficiently.

Meeting the Build Criteria

To effectively meet the *"What You Must Build"* criteria, the solution was carefully designed around the capabilities of **Google ADK (Agent Development Kit)**.

The development process involved:

- Designing a **custom AI agent architecture** using ADK to handle input, processing, and output flow
- Implementing **HTTP-based communication**, allowing external systems to interact with the agent
- Integrating **Google Gemini API** to enable intelligent query understanding and response generation
- Defining a **structured response format** to ensure consistency and usability

The system was further:

- **Containerized using Docker**, ensuring portability and environment consistency
- **Deployed on Google Cloud Run**, enabling scalability and public accessibility

By combining ADK, Gemini, and Cloud Run, the solution satisfies all key requirements:

- ADK-based agent design
- Single-task functional implementation
- API-accessible system
- Cloud deployment

This demonstrates a complete end-to-end AI solution aligned with the track requirements.

Opportunities

Difference from Existing Solutions

- Focuses on a **single-task AI agent** instead of complex chatbots
- Provides **structured and concise outputs**
- Designed for **API-based usage**, not just conversation

Problem Solving Approach

- Delivers **fast and relevant information**
- Reduces unnecessary long responses
- Ensures **consistent and usable output format**

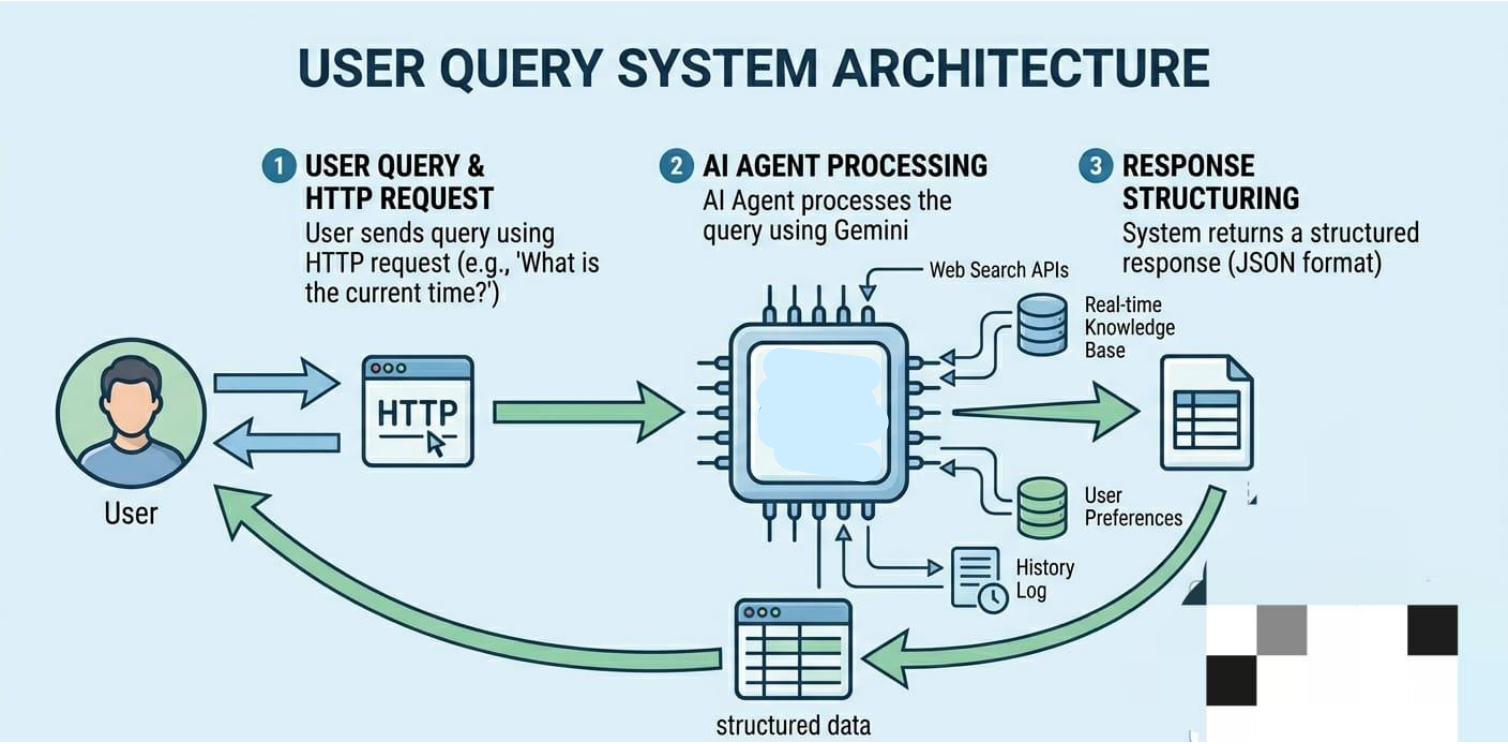
USP (Unique Selling Point)

- Lightweight and efficient architecture
- Easy to deploy and scale (Cloud Run)
- Clean integration using ADK + Gemini
- Structured responses for better usability

List of features offered by the solution

- Accepts user queries via REST API
- Processes natural language input using Gemini
- Generates **structured, concise, and consistent responses**
- Lightweight and efficient execution
- Modular architecture using ADK
- Easily deployable on cloud infrastructure
- Scalable for multiple users and requests
- Minimal latency response handling
- Can be extended to domain-specific applications

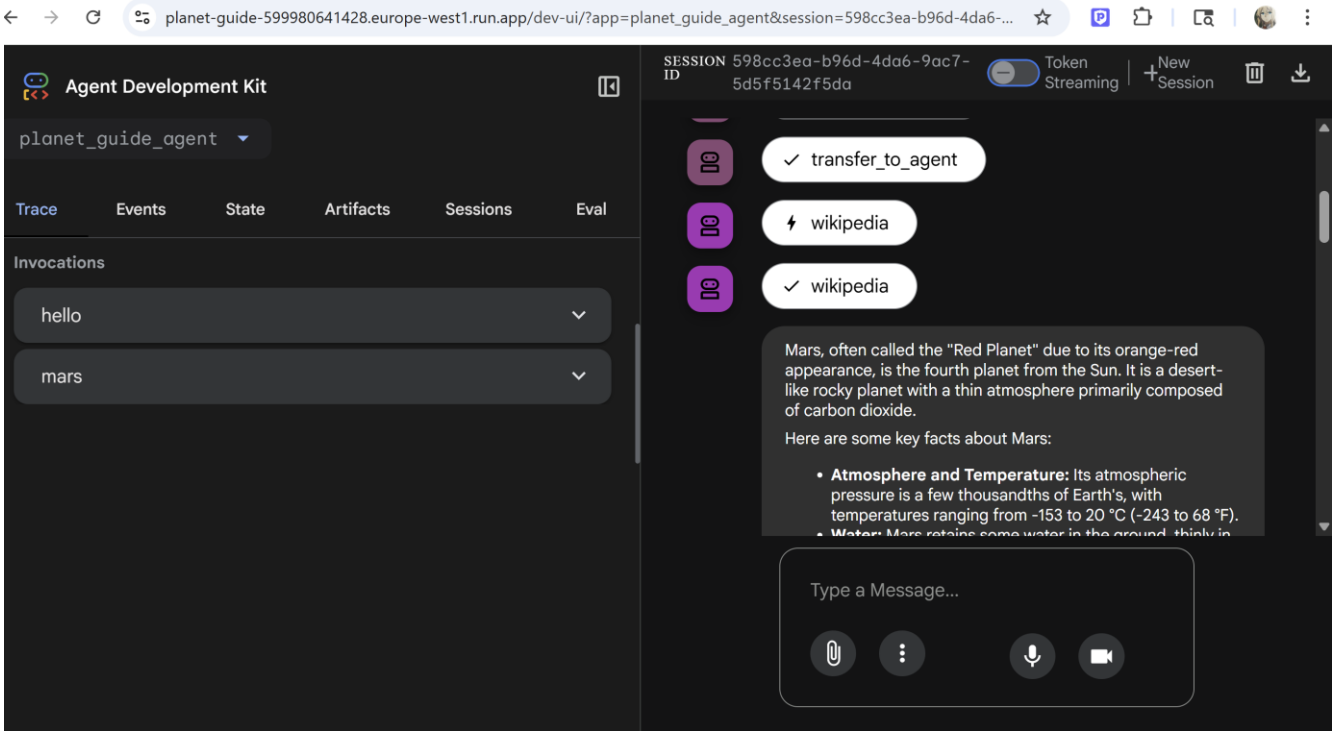
Process flow diagram



Technologies to be used in the solution

- **Google ADK (Agent Development Kit)**
  Used to design and manage the AI agent workflow and structure
- **Google Gemini API**
  Provides natural language understanding and intelligent response generation
- **Python**
  Used for implementing backend logic and agent behavior
- **Docker**
  Enables containerization for consistent deployment environments
- **Google Cloud Run**
  Provides scalable, serverless deployment of the AI agent
- **REST API**
  Facilitates communication between the user and the AI system

Snapshots of the prototype



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Thank you